



Data Analytics Capstone Topic

Final Presentation

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Executive Summary

- **Objective:** Forecast next-quarter Midcap vs NIFTY excess return using monsoon rainfall + economic signals.
- **Why it matters:** Weather and macro surprises often move markets; we test whether combining them improves timing for midcap exposure.
- **Data:** India monsoon deviation from normal (IMD), inflation, GDP, policy rate, and index returns (2013–2025), aligned by quarter.
- **Approach:** Compare a simple baseline (previous quarter's return) to an expanded model (adds monsoon + macro). Walk-forward validation.
- **Result:** No model beats naïve baselines on this sample. Recent relative performance (last quarter's spread) carries the most usable signal. Macro variables (inflation, GDP, policy rate, monsoon) show weak, unstable lift at a quarterly horizon



Gap Analysis

- Prior work links monsoon → GDP and macro → equity returns separately.
- Gap: No applied business study combining monsoon + macro to forecast midcap excess return next quarter.
- Business use: If a “good monsoon + supportive macro” reliably preceded stronger midcap quarters, allocators could tilt exposure tactically; policy teams could quantify climate sensitivity.



Research Questions

1) Does deviation in monsoon rainfall significantly correlate with Midcap breakouts?

Method: Pearson/Spearman; 12-Q rolling; partial correlations (trend/seasonality); lagged cross-correlation.

2) Which macroeconomic indicators best predict Midcap 100 breakouts? (considered: GDP growth YoY, policy-rate change in bps)

Method: Expanding-window CV (5 folds); compare Ridge, Lasso, GBR, RF vs naïve baselines; selection by RMSE.

3) Does a combined monsoon-macroeconomic model outperform models using single-signal inputs?

Method: Ablation in same CV (momentum-only, monsoon-only, macro-only, combined); compare RMSE.

4) Which features (e.g., lagged monsoon, repo-rate changes, GDP) have the highest predictive weight?

Method: Permutation importance on small holdout; drop-one ablation; L1/L2 sanity checks; partial correlations.



Data Description and EDA

Sources & scope

- Rainfall: IMD, All-India monsoon deviation from normal (Jun–Sep), 2012–2025.
- Gross Domestic Product(GDP): MoSPI(Institute), quarterly GDP growth (YoY, %), 2014–2025.
- Policy rate: RBI, repo rate level and quarterly change (basis points), 2014–2025.
- Indices: NSE Indices, NIFTY 50 & NIFTY Midcap 100 daily levels → quarterly returns, 2013–2025.

Frequency & alignment

- All inputs aligned to calendar quarter-end.
- Monsoon deviation stamped at Sep-30 each year and forward-filled until the next monsoon season.
- Target is constructed on quarterly data.

Target variable

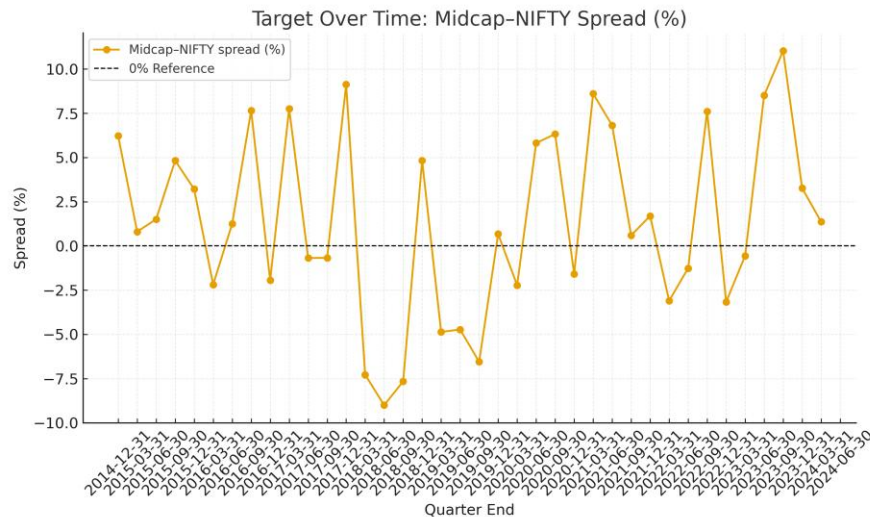
- Next-quarter Midcap vs NIFTY outperformance (%)
(i.e., Midcap quarterly return minus NIFTY quarterly return, predicted one quarter ahead.)

Features used at time T

- Monsoon deviation (%), GDP growth (YoY, %), Policy rate change (bps),
- Momentum: last-quarter spread & 4-quarter average,
- Seasonality: quarter-of-year (sine/cosine).

Note: CPI was evaluated in the interim study and excluded from the final model due to very low incremental signal.

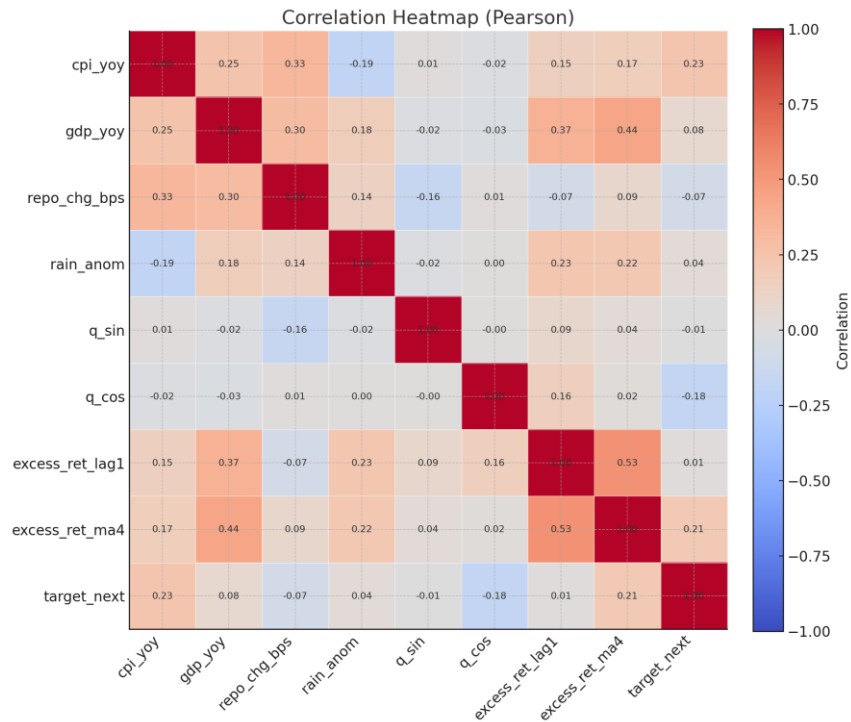
Data Description and EDA



- The Midcap-NIFTY spread is the quarterly excess return of the Midcap index over the NIFTY benchmark.
- This is the target variable we are modeling.
- The chart shows historical spreads by quarter (positive = Midcaps outperform NIFTY, negative = underperform).
- A 0% reference line is added for clarity:
- Above line → Midcaps outperform
- Below line → NIFTY outperforms
- **Takeaway:** The spread exhibits clear fluctuations over time, supporting the need for predictive modeling.

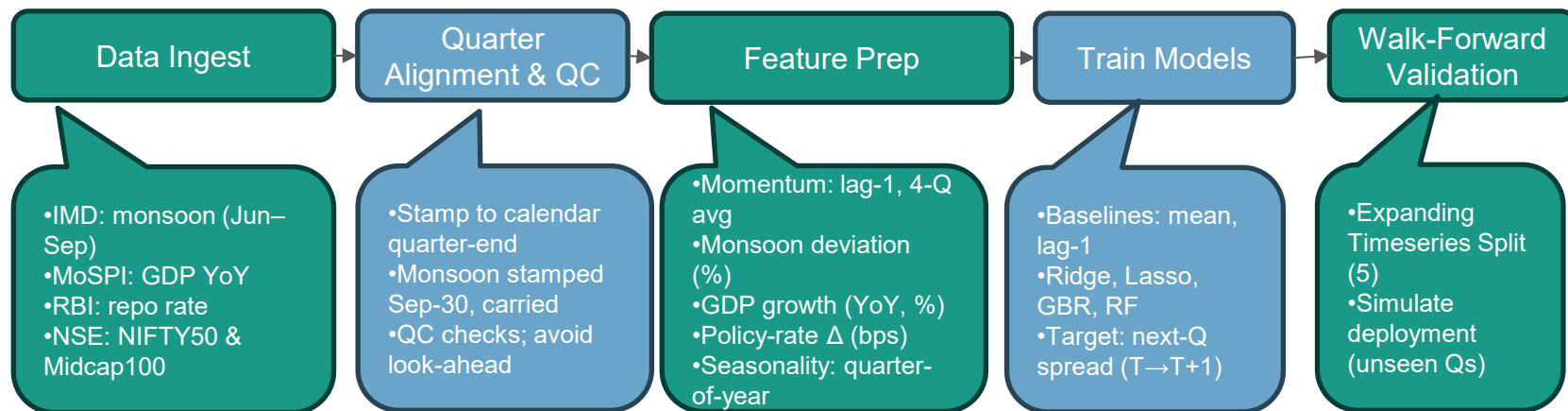
Data Description and EDA

- Quarterly Pearson correlations. Most relationships are weak; the strongest with the target are still small ($|r| \lesssim 0.23$).
- Macro variables are modestly correlated with each other but not with next-quarter outperformance.





Architecture diagram/Workflow





Model building

Target (what we predict):

Next-quarter Midcap – NIFTY spread (%).

Features (available at time T):

- Monsoon deviation (%) (IMD, Jun–Sep stamped Sep-30 and carried)
- GDP growth (YoY, %); policy-rate change (bps)
- Momentum: last-quarter spread; 4-quarter average
- Seasonality: quarter-of-year (sin/cos)

Models compared:

Ridge, Lasso, Gradient Boosting, Random Forest vs naïve baselines (train-mean; lag-1 only).

Validation design:

Expanding TimeSeries Split (5 folds), no shuffling; features at T, target at T+1 to avoid look-ahead.

Evaluation:

primary RMSE; report MAE, R^2 as supporting metrics.



Results

Model Performance Comparison						
Rank	Model	Type	RMSE	MAE	R ²	Δ vs Best Baseline (RMSE)
1	Train-mean	Baseline	0.063	0.053	-0.286	+0.000
2	Lag-1 only	Baseline	0.066	0.054	-0.396	+0.003
3	Random Forest	Model	0.079	0.065	-0.975	+0.016
4	Lasso	Model	0.083	0.066	-1.190	+0.020
5	Gradient Boosting	Model	0.087	0.074	-1.430	+0.024
6	Ridge	Model	0.146	0.105	-5.830	+0.083

- On this dataset, simpler wins: the baseline makes more correct up/down calls
- Adding a rainfall-based GDP improved the expanded model slightly, but still under the baseline.

Results

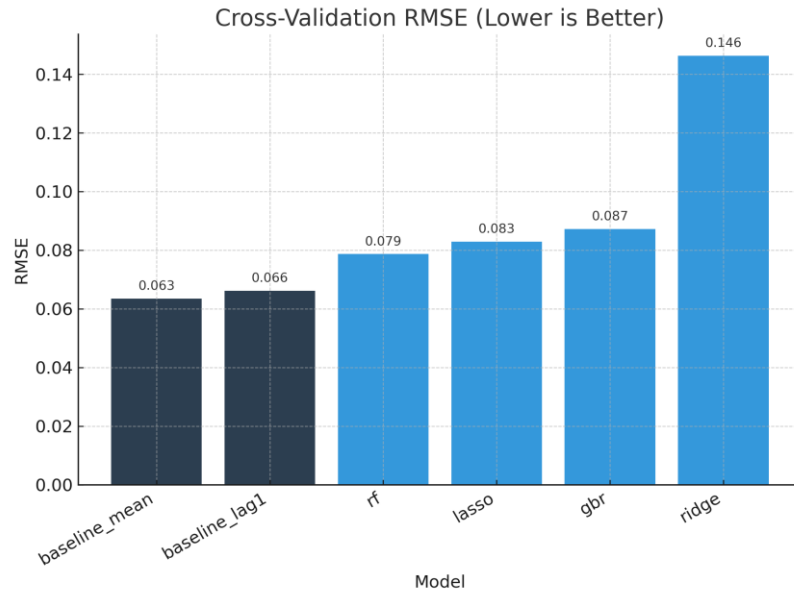
Results – Cross-validation RMSE

Why it matters:

- RMSE is the main accuracy metric (lower = better).
- Benchmarks the models against simple baselines.

Key Findings:

- Baselines (mean & lag-1) achieve the lowest RMSE.
- All machine learning models (RF, Lasso, Gradient Boosting, Ridge) fail to beat baselines.
- Indicates that macro signals add little value beyond naïve baselines.



Results

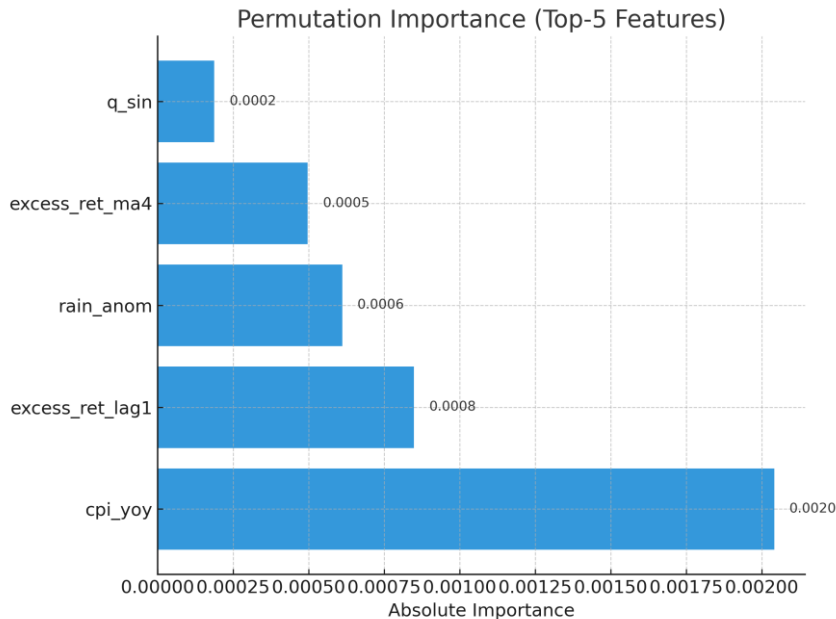
Interpretation – Permutation Importance

Why it matters:

- Identifies which features contribute most (even if weakly) to predictions.
- Helps assess whether macro variables add value.

Key Findings:

- Momentum signals dominate lagged excess return and moving average show the most predictive value.
- Macro variables are weak: CPI and rainfall anomalies add very little explanatory power.
- Seasonal terms (q_sin, q_cos) are negligible.



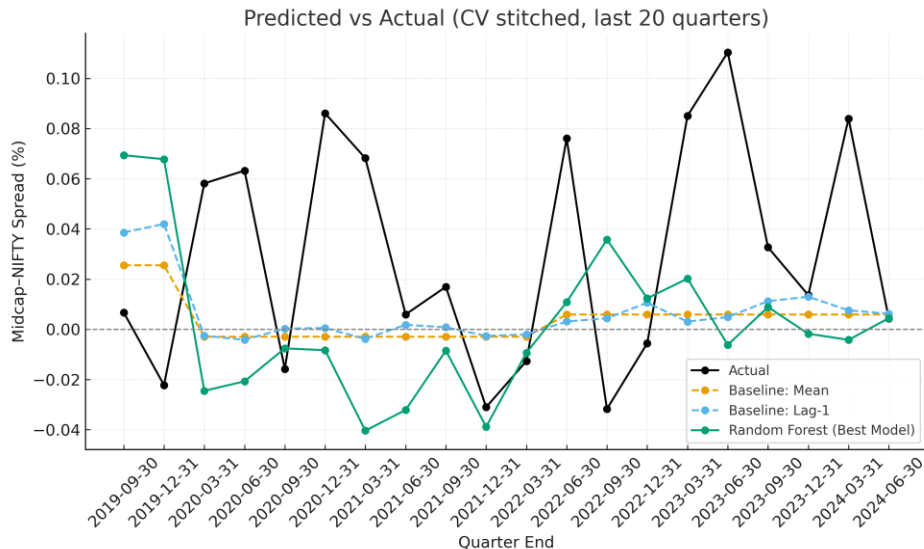
Results

Why it matters:

- Visual check of how well models track the true Midcap-NIFTY spread.
- Highlights **tracking error** (gaps between actual and predictions).

Key Findings:

- Baselines (mean, lag-1) generally track the spread as well as, or better than, models.
- The “best” model (Random Forest) shows **visible error and lag** vs actuals.
- Confirms that machine learning models provide **no improvement over baselines**.





Results

Interpretation:

- Quarterly macro $\neq$ strong edge: CPI, GDP, repo, and monsoon don't add reliable power vs naïve yardsticks.
- Momentum carries the signal, but not enough to beat a constant mean on error.
- Why: Quarterly aggregation blurs timing; effects likely priced or acting at different horizons; small sample (~ 39) $\rightarrow$ high variance.



Implementation and User Benefit

- For portfolio teams: Keep the baseline as the production default; treat inflation as a light overlay signal (guardrails on position size).
- For sell-side: Publish a quarterly “Monsoon + Macro” note translating the latest IMD + CPI prints into scenario commentary.
- For policy: Use the framework to discuss climate → growth → equity sensitivity; monitor whether the relationship strengthens with more data.



Conclusion

- Conclusion: With the current sample, a simple baseline beats more complex variants. Inflation (YoY) shows the only consistent informational edge; monsoon effects are directional but weak.
- Limitations:
 - Sample size (~39 trainable quarters) → unstable estimates; negative R^2 vs naïve baselines.
 - Aggregation & timing: Monsoon stamped at Sep-end and carried; GDP/CPI as YoY quarterly figures; repo moves are stepwise; publication lags matter.
 - Scope: Relative index spread only; no costs/portfolio construction.
- Next steps: Backfill more history, test regime-aware models (e.g., “high-inflation vs low-inflation” regimes), and explore sector-level effects to increase sample size.



Bibliography

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- Ministry of Statistics & Programme Implementation. (n.d.). Consumer Price Index (Base 2012=100). Retrieved September 7, 2025.
- National Statistical Office, MoSPI. (n.d.). Quarterly estimates of GDP at constant (2011–12) prices. Retrieved September 7, 2025.
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- NSE Indices Limited. (n.d.). NIFTY 50; NIFTY Midcap 100 — historical data. Retrieved September 7, 2025.
- (Optional) Academic articles exploring monsoon–GDP and macro–equity relationships (list in notes with year, title, journal, DOI).